

PODGORICA GOLUBOVCI, Montenegro

WMO: 134620

Lat: 42.367N

Lon: 19.250E

Elev: 33

StdP: 100.93

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.7	-1.3	-13.3	1.2	1.9	-10.8	1.5	3.0	13.4	2.6	12.1	4.0	2.9	0	0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	36.3	21.4	34.9	21.2	33.5	21.0	23.3	31.9	22.5	31.4	21.8	30.8	3.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.6	15.4	27.1	19.8	14.6	25.7	19.0	13.9	25.3	69.2	31.8	66.2	31.3	63.8	31.0	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	8.9	7.3	-5.4	38.3	1.7	2.0	-6.6	39.7	-7.6	40.9	-8.5	42.0	-9.8	43.5		
(5)			-6.6	24.9	2.0	1.5	-8.0	26.0	-9.2	26.9	-10.3	27.8	-11.8	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	5.7	7.2	10.8	15.0	19.4	23.8	27.1	27.2	21.7	16.7	11.3	6.9		
	DBStd	8.11	3.12	3.40	3.21	3.00	3.06	3.32	2.73	2.86	3.06	2.97	3.37	3.19		
	HDD10.0	386	137	88	28	2	0	0	0	0	0	1	25	105		
	HDD18.3	1708	392	313	232	105	25	3	0	0	6	66	211	355		
	CDD10.0	2622	3	9	54	153	291	415	531	533	352	208	64	9		
	CDD18.3	902	0	0	0	6	57	168	272	276	108	16	0	0		
	CDH23.3	9473	0	1	1	54	438	1704	3255	3157	781	82	1	0		
	CDH26.7	4263	0	0	0	8	99	689	1629	1583	248	6	0	0		
(14) Wind	WSAvg	2.4	2.4	2.7	3.0	2.3	2.4	2.3	2.6	2.4	2.3	2.0	2.0	2.5		
(15) Precipitation	PrecAvg	1707	170	158	143	147	90	57	46	66	135	178	242	217		
	PrecMax	2359	381	385	349	340	230	162	140	274	390	523	639	413		
	PrecMin	987	0	0	6	14	0	8	0	5	0	0	20	36		
	PrecStd	311	114	99	81	75	55	40	30	61	105	124	122	107		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	18.3	22.6	27.6	30.8	35.5	37.9	38.7	34.1	27.4	21.5	16.3		
		MCWB	11.3	11.6	12.9	16.5	19.4	22.1	21.6	21.8	20.3	18.4	15.7	13.0		
	2%	DB	13.3	15.9	20.1	24.8	29.2	33.8	36.3	36.7	31.5	25.5	19.5	14.7		
		MCWB	10.0	10.6	12.3	15.4	19.0	21.8	21.5	21.5	19.5	17.5	15.0	12.0		
	5%	DB	11.9	14.3	18.3	22.8	27.7	32.2	34.9	35.1	29.4	23.9	18.1	13.5		
		MCWB	8.9	9.8	11.5	14.3	18.3	21.2	21.3	21.0	19.1	16.8	14.0	10.8		
	10%	DB	10.6	12.7	16.6	21.0	25.9	30.5	33.5	33.6	27.7	22.3	16.6	12.1		
		MCWB	8.1	8.9	10.6	13.8	17.7	20.6	21.0	20.8	18.7	16.3	13.3	9.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	13.4	15.9	17.5	20.9	24.4	24.7	24.3	22.1	20.2	17.2	14.6		
		MCDB	14.0	15.8	18.4	25.1	28.9	32.3	33.0	33.0	28.5	24.5	19.3	15.3		
	2%	WB	11.1	11.8	13.7	16.4	19.7	22.9	23.2	23.1	21.2	19.1	16.1	13.0		
		MCDB	12.5	14.6	17.5	23.1	27.1	31.5	31.9	31.9	27.4	22.9	18.5	14.2		
	5%	WB	9.9	10.8	12.6	15.5	18.9	22.0	22.4	22.3	20.5	18.1	15.0	11.7		
		MCDB	11.0	13.2	16.2	20.8	26.1	30.6	31.7	31.3	26.5	21.6	17.3	12.8		
	10%	WB	8.9	9.8	11.7	14.7	18.1	21.1	21.8	21.6	19.7	17.3	13.9	10.2		
		MCDB	10.0	12.0	14.9	19.3	24.6	29.1	31.2	30.7	25.5	20.5	15.9	11.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	8.0	8.4	9.1	9.6	10.2	10.7	10.8	9.4	9.1	8.3	7.5		
		MCDBR	9.4	11.0	11.6	12.4	12.1	11.9	12.6	12.6	11.5	11.3	10.1	9.0		
	5% WB	MCWBR	6.2	6.5	5.9	5.4	4.9	4.6	4.2	4.1	4.4	5.3	5.8	5.9		
		MCWBR	6.9	8.5	9.4	10.8	11.1	11.4	11.4	11.5	10.0	9.0	8.2	5.9		
(40) Clear-Sky Solar Irradiance	taub		0.339	0.365	0.415	0.471	0.460	0.442	0.443	0.454	0.431	0.398	0.368	0.339		
		taud	2.392	2.294	2.183	2.056	2.153	2.258	2.251	2.213	2.263	2.334	2.369	2.402		
	Ebn at Noon		791	824	821	800	821	836	830	807	797	778	747	755		
		Edh at Noon	81	104	131	160	149	135	134	135	120	99	83	74		
(44) All-Sky Solar Radiation	RadAvg		1.63	2.36	3.51	4.60	5.73	6.65	6.87	6.13	4.37	3.03	1.90	1.46		
	RadStd		0.19	0.36	0.50	0.51	0.47	0.46	0.39	0.41	0.36	0.32	0.20	0.27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	-102	N/A	N/A

Nomenclature: See separate page